Card fraud triage

Gaussian mixture density scoring

Phase 1: Machine Learning

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The review task

Learn normal activity from earlier legitimate transactions.

Give less typical transactions a larger anomaly score.

Measure fraud found and legitimate transactions sent for review.

Phase 1 model: GMM. Autoencoder comparison belongs to Phase 2.

Literature behind our choices

Source	Idea	Use in our project
Dempster et al., 1977	EM estimates hidden memberships	Fit the Gaussian components
Saito & Rehmsmeier, 2015	PR reveals rare-class alert burden	Lead with PR and review metrics
scikit-learn mixture guide	Covariance and information criteria	Diagonal GMM, AIC and BIC

Our contribution is a clear, reproducible baseline experiment.

Dataset quality

Measured quantity	Original CSV	After exact-repeat removal
Transactions	284,807	283,726
Fraud labels	492	473
Exact repeats removed	1,081	19 of these carry fraud labels

31 columns. No missing or infinite cells. Source checksum recorded.

V1-V28 are anonymous PCA inputs. Business meanings are unavailable.

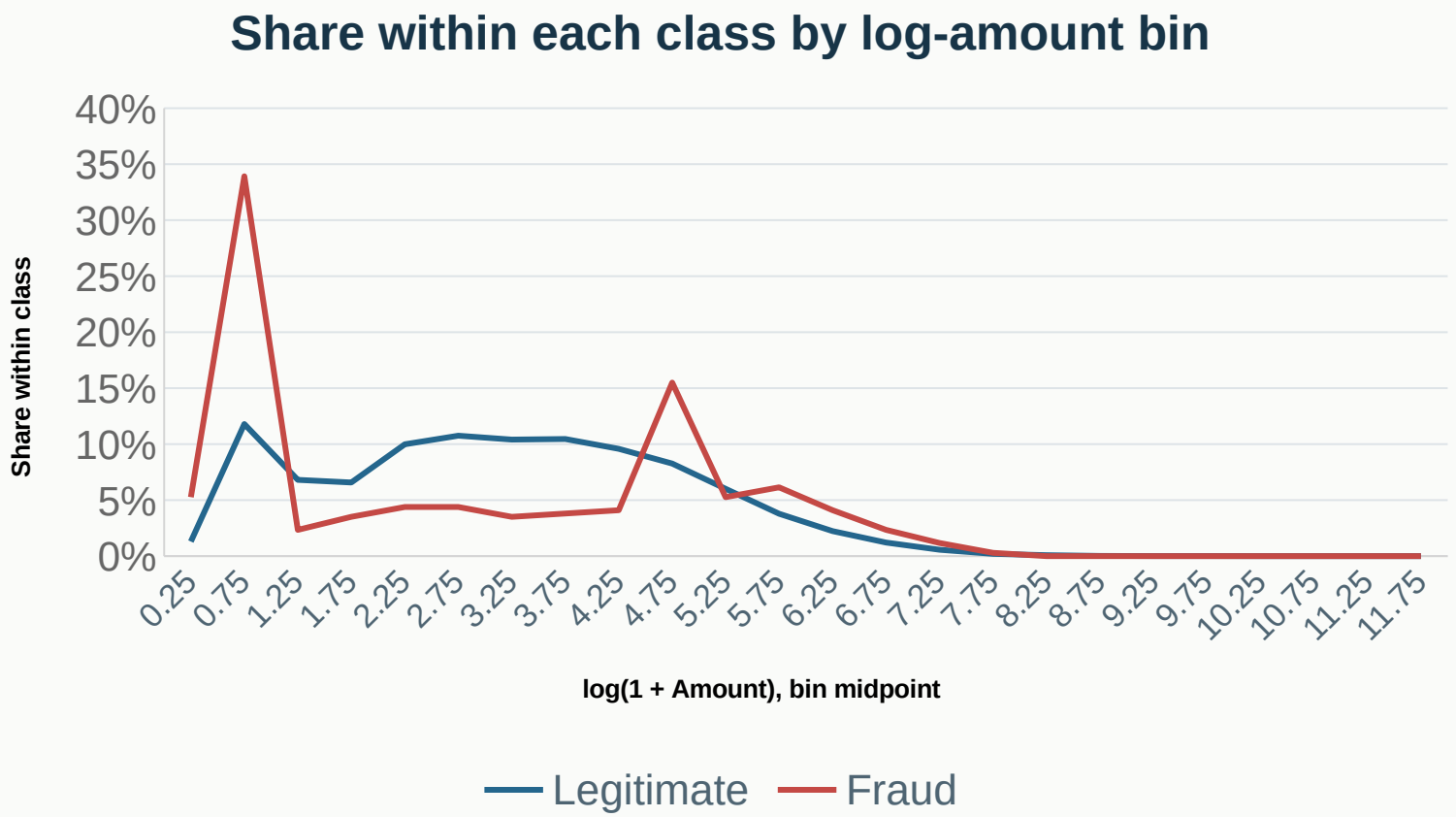
Chronological evaluation

Period	Rows	Legitimate	Fraud
train	170,235	169,893	342
validation	56,745	56,688	57
test	56,746	56,672	74

Fit on 169,893 legitimate training rows. Freeze choices before test.

Equal timestamps stay together. No future rows fit preprocessing.

Training amount distribution



Amount spans a wide range.

The log transform compresses the tail and handles zero.

The 29 model inputs

V1-V28 plus LogAmount = log(1 + Amount)

Time orders the split. Class filters legitimate history and evaluates results.

z = (value - legitimate training mean) / legitimate training standard deviation

Validation and test use these saved training values.

Class and Time never enter the model feature matrix.

How the GMM learns

p(x) = sum of [component weight x Gaussian density]

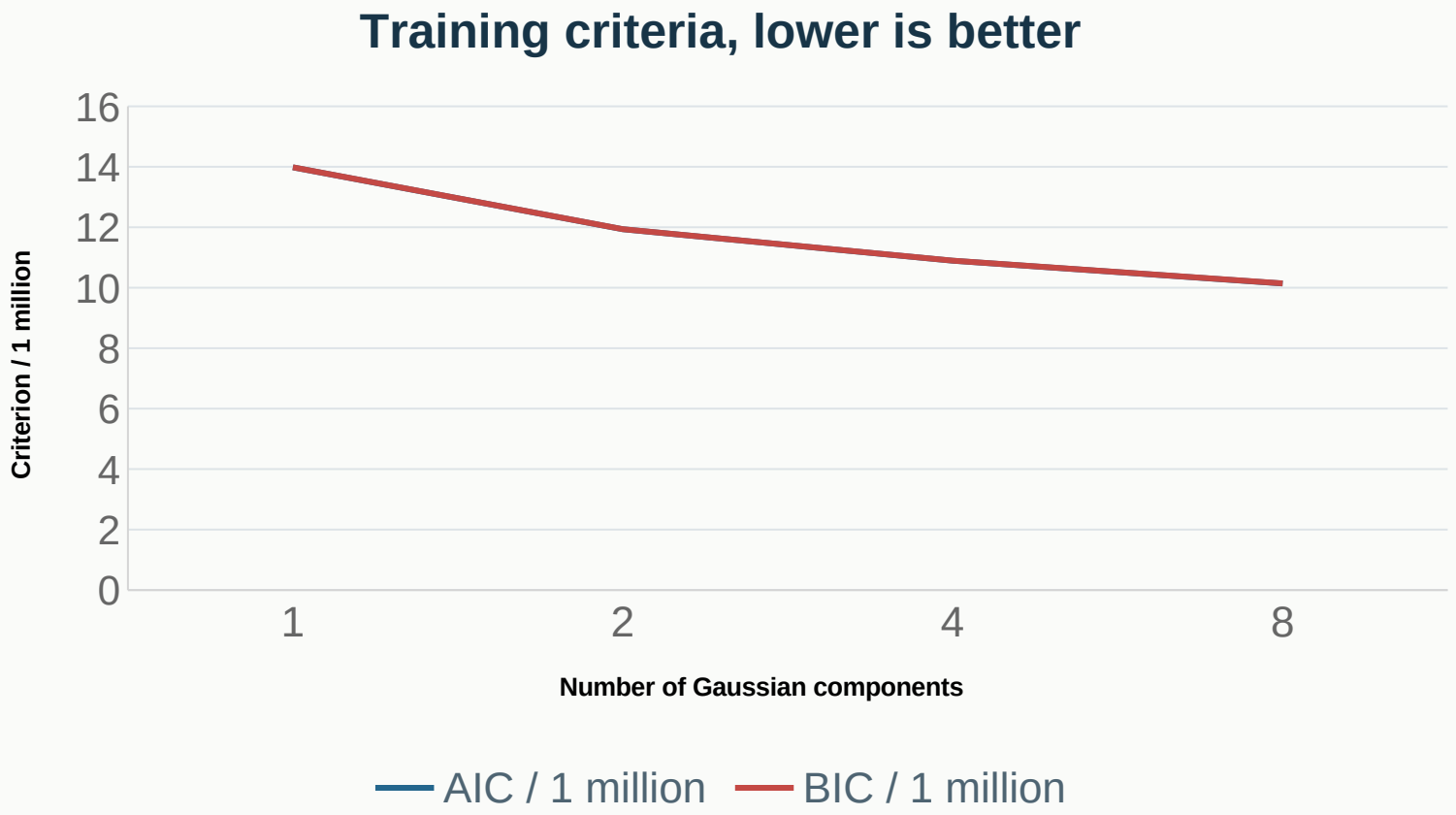
E-step: estimate soft membership in each Gaussian.

M-step: update weights, centers and diagonal variances.

Three starts. Seed 42. Up to 200 iterations. Small variance regularization.

Anomaly score = -log p(x). Larger means less typical of legitimate activity.

Component selection

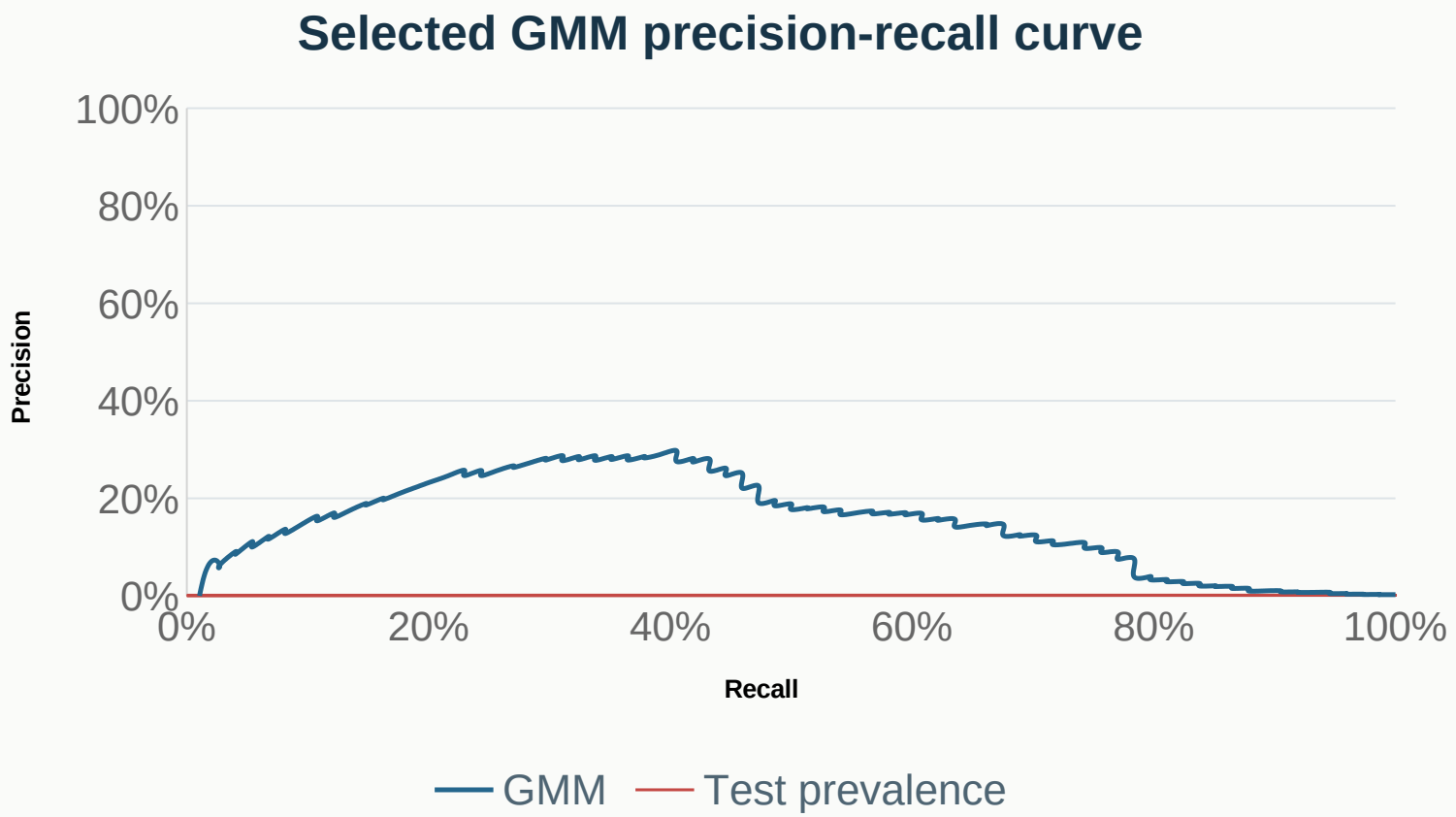


K = 8 wins in our grid.

All candidates converge.

Eight is the largest value tested.

Final test ranking



Average precision
GMM: 0.1532
One Gaussian: 0.0540

PR AUC: 0.1496
Test fraud: 0.1304%

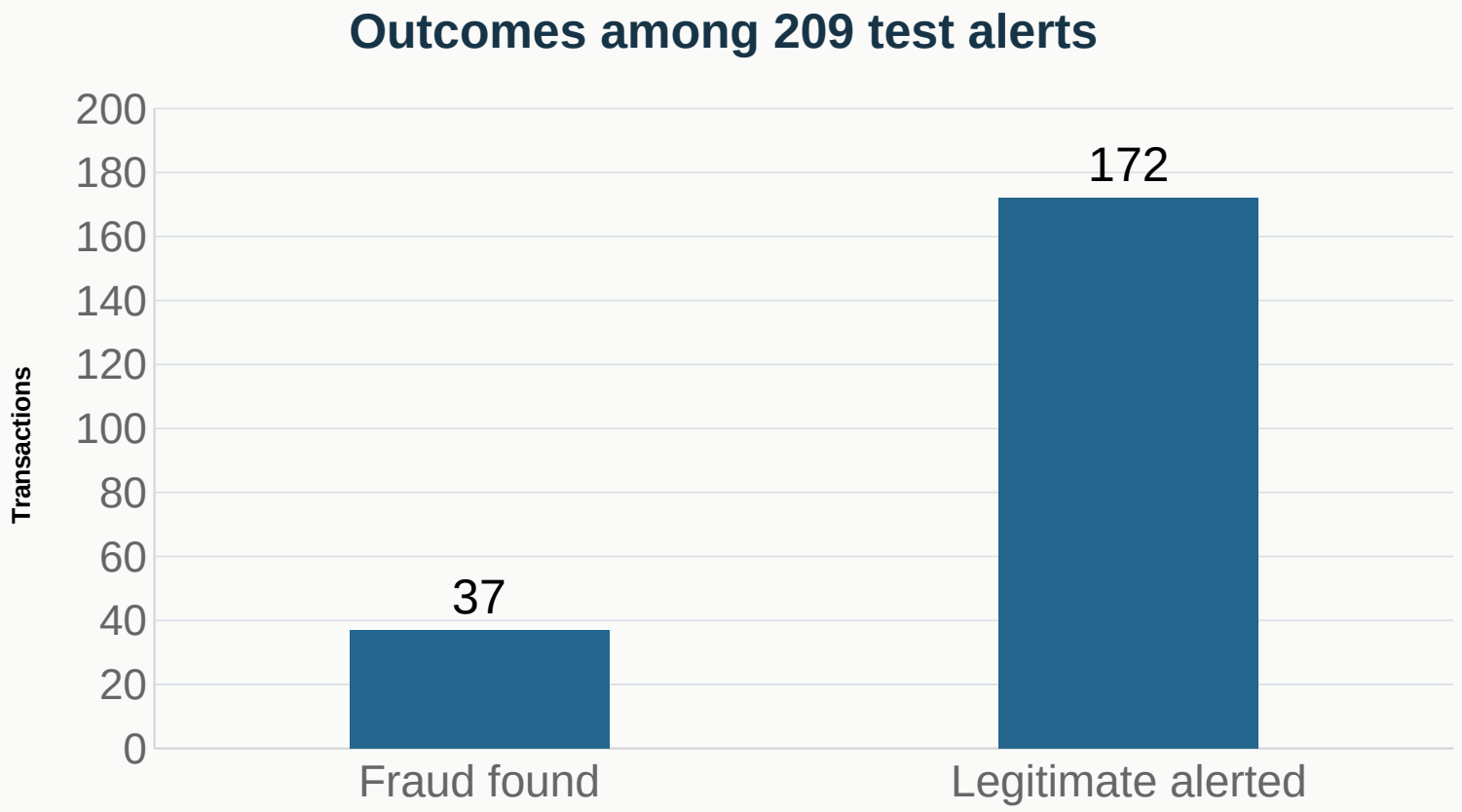
Validation policies on the test period

Validation budget	Fraud found / 74	False alerts	Test FPR
0.1%	6	39	0.0688%
0.5%	37	172	0.3035%
1.0%	52	381	0.6723%

Default 0.5% policy: threshold 75.8942, with score > threshold.

A validation budget may give a different false-positive rate later.

The default review workload

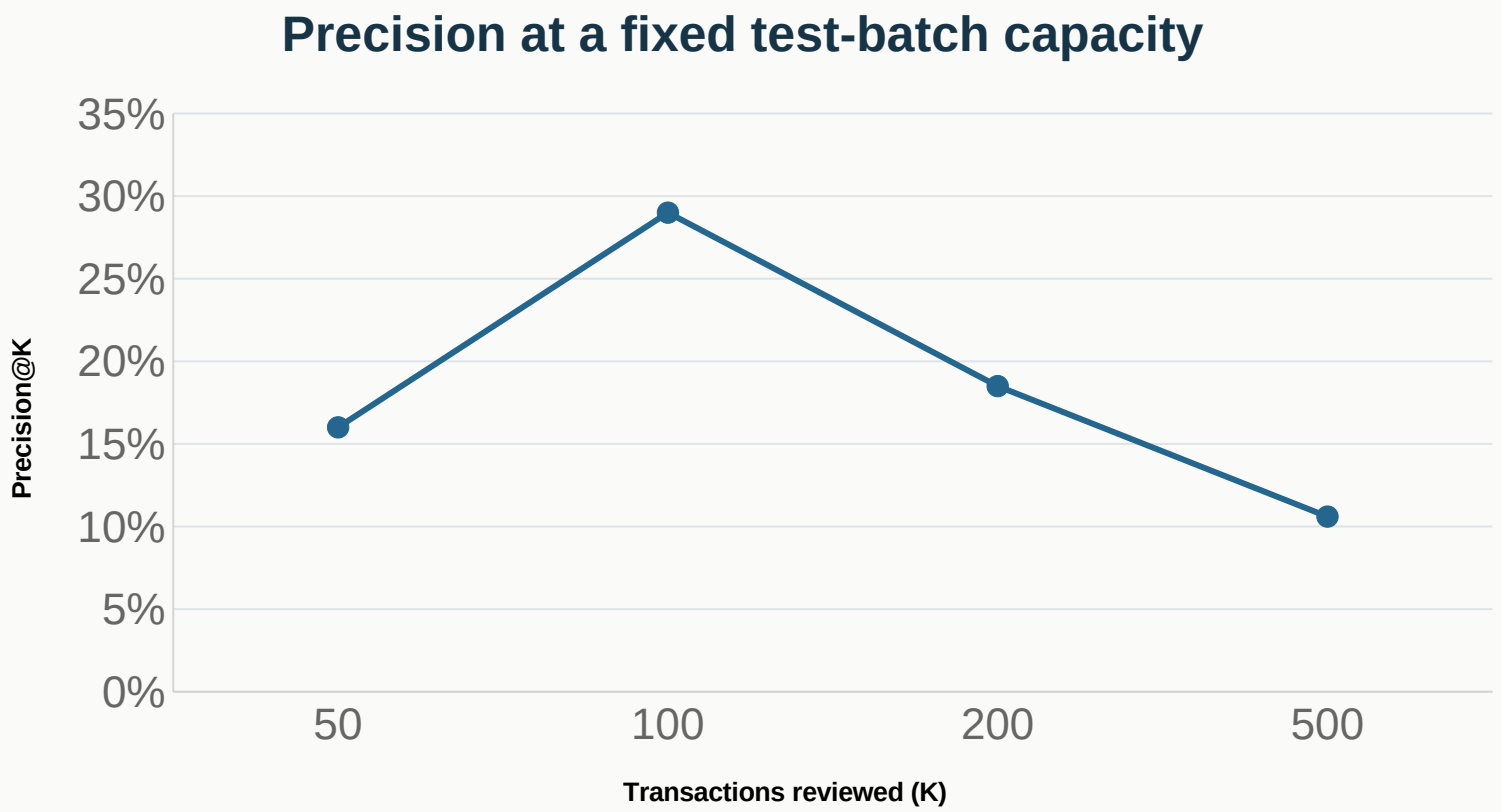


Recall: 37 / 74 = 50%

Precision: 37 / 209 = 17.70%

Missed frauds: 37

Fixed batch review capacity



Top 100 reviews

29 frauds found

29% precision
39.19% recall

Limits and Phase 1 conclusion

The GMM improves this ranking over one Gaussian.

At the default policy, it catches half the test fraud and creates 172 false alerts.

Only 74 test frauds. Anonymous inputs. Short history. Restrictive density assumptions.

Phase 2: compare an autoencoder under the same data and evaluation protocol.

Every member should explain the split, score, threshold and exact metric counts.